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Development and validation of a polyfit approach for assessing alveolar bone loss using panoramic radiography

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Abstract

Background Panoramic radiographs (PAN) are one of the most common diagnostic tools in clinical practice. Periodontal disease, the second most prevalent oral disease, significantly impacts patients' quality of life. However, there is currently no standardized and quantitative image analysis method for periodontal diagnosis. This study aims to estimate alveolar bone loss in six sextants of the mouth using the Polyfit approach on panoramic radiography. This approach utilizes ratio and proportional measurements based on fixed anatomical points to improve the accuracy of assessing bone loss.

Methods In this retrospective clinical study, we assessed alveolar bone loss (ABL) in 290 subjects. The subjects were divided into two groups, the resorption group (abbreviated to ABL) and non-resorption group (abbreviated to non-ABL), based on the presence of any ABL sextants. Each tooth was manually marked with reference anatomical landmarks using Anaconda-Labelme 5.2.1 software. To evaluate the proportionate bone resorption for each tooth, we employed the PAN-POL model based on the polyfit function to quantify bone loss across six dental sextants and the entire dental arch. For the reliability of measurements, Cone Beam Computed Tomography (CBCT) data from 30 patients were selected. Measurements were conducted in Mimics 21.0 and compared with the model's results to validate the ABL assessment. An independent sample t-test or the intergroup rank-sum test was used to evaluate the difference between resorption data from both classification methods, and ABL and non-ABL groups in each classification. Pearson's correlation analysis and linear regression analysis were used to test and verify the correlation between CBCT and panoramic radiography. *P* values of > 0.05 were considered not statistically significant.

Results The PAN-POL model demonstrated effective differentiation between resorption and non-resorption groups, and the groups based on two classification achieving statistical significance ($P < 0.05$), and the ratio results were consistent with those from CBCT ($P > 0.05$), indicating no significant difference. The Intraclass Correlation Coefficient

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(ICC) results for reliability testing among two experts for both PAN and CBCT were steadily 0.83 ± 0.06 and 0.93 ± 0.06 ($ICC > 0.75$).

Conclusion In this study, the PAN-POL model accurately measured ABL in panoramic images by incorporating standard anatomical landmarks. This model aids in Periodontal Screening and Recording, serving as a novel, valuable and intuitive tool for initial periodontal diagnosis.

Keywords Periodontitis, Alveolar bone loss, Panoramic radiography, Diagnostic imaging

Introduction

Periodontal disease (PD) is recognized as the third most pervasive chronic health risk, succeeding malignant neoplasms and cardiovascular disorders [1]. With a global prevalence estimated between 20 and 50%, it is classified within the top six chronic morbidities [2]. PD is also the predominant etiologic agent of dentition loss in the adult demographic [3]. The diagnosis of periodontal conditions, notably periodontitis, is established through a combination of clinical examinations and imaging studies. Clinical assessments involve the measurement of gingival bleeding on probing (BoP%) [4], periodontal probing pocket depth (PPD) and clinical attachment loss (CAL) [5]. These parameters are essential for evaluating the integrity of the alveolar bone's coverage around the tooth root.

Estimation the extent alveolar bone loss (ABL), the most striking feature about periodontitis, is based on imaging diagnostic radiography [6–8]. Previous studies advocate to use the periapical radiograph [9, 10] and cone-beam computed tomography (CBCT) [11, 12] for individual teeth and their adjacent structures diagnoses, otherwise exclude panoramic radiographs (PAN) due to concrete artifact that may arise [12, 13]. These artifacts can arise from various sources, such as patient movement, positioning errors, or overlapping anatomical structures, leading to distortions and inaccuracies in the images. As a result, while PAN is considered a most useful as well as affordable clinical tool for initial assessments due to the general information it presents and its ability to reduce radiation exposure [14–16], offering a comprehensive view of the whole maxillofacial region, containing mandibular and maxillary dental arches and their supporting structures in a single image, its limitations necessitate the use of more precise approach through this imaging techniques. Thus, enhancing the precision of information extracted from panoramic radiography holds significant clinical value, especially when it comes to an economical and practical screening method of various oral and maxillofacial diseases, including periodontitis, dental caries, and temporomandibular joint disorder [17].

According to the current classification of periodontal and peri-implant diseases and conditions developed in 2017, Stage I and II periodontitis are associated with the

presence of horizontal bone loss, while the presence of vertical bone losses and furcation defects in Stage III and IV periodontitis is noted [18, 19]. As for the patients distribution and resorption pattern, patients with horizontal bone loss, which is resorbed positioned more apically to the parallel line that should be in the healthy periodontium, present a larger proportion [20]. And it always occurs when the bone supporting the tooth melts at the same levels on all tooth-related surfaces (mesial, distal, vestibular/buccal, lingual/palatal) in adjacent neighbor teeth and periodontal tissue [21]. This condition is commonly seen in the area centered on the first molars, upper and lower central incisors, which can be considered as regions segmented by the distal face of the canine [22]. Therefore, the more specific and quantitative analysis related to dental arch segmentation and ABL visualization is necessary.

Nevertheless, the rotational nature of panoramic imaging can introduce artifacts and lead to the image distortion and magnification, especially concerning the accurate measurement of root length, which is crucial for precise periodontal diagnostic evaluation [23]. While the transition at the mouth corners is visible, the features of panoramic imaging strengthen the significance of partition analysis.

Several studies carry out the proportion index related to tooth and periodontal tissue structure to avoid the image distortion, including root-crown ratio (or crown-root ratio) [24, 25] and periodontal index [26], providing a more precise blueprint for dental assessment. Meanwhile, a multitude of global studies previously describe various techniques for assessing ABL on radiography, utilizing tools such as compasses [27], rulers [28, 29], millimeter probes [30], grid patterns [31], and software [8, 32]. However, Fukuda et al. [33] highlight the absence of a proportionate radiographic assessment for consecutive quantitative bone levels instead of scoring, leading to the inconclusive diagnosis and rough staging/grading standards when it comes to radiographical periodontal scoring in clinical practice among young dentists or non-periodontal specialists.

Previously, two representative scores developed by the World Health Organization (WHO) and the American Dental Association (ADA) for evaluating periodontal status by dividing the mouth into sextants through

screening are Community Periodontal Index of Treatment Needs (CPITN) [34] and Periodontal Screening and Recording (PSR) [35]. Both systems provide a standardized way to quickly assess periodontal health in clinical examination, with CPITN focusing on treatment needs at a population level and PSR serving as a screening tool in clinical practice. However, there has been still a lack of consensus on the optimal method for radiographic evaluation of quantitative bone levels inside the sextant section in imaging diagnosis [14, 36], considering typical diagnosis of a single tooth measurement is relatively expensive, time consuming, and subjective scoring.

Built on the polyfit function and simple ratio, this study proposes a new mathematical approach to estimate alveolar bone loss in sextants using panoramic radiographs in a more anatomical and visible presentation to assist clinical diagnosis Fig. 1.

Methodology

This clinical retrospective investigation was conducted from January 2022 to December 2023 in West China Hospital of Stomatology, Sichuan University. For this investigation, the sample comprised two imaging modalities: Orthopantomography (PAN, $n_1 = 290$) and with corresponding Cone Beam Computed Tomography (CBCT, $n_2 = 30$). The research was in conformity with the guidance of the World Medical Association Helsinki Declaration for biomedical research involving human subjects and the imaging data involved was approved by the institutional review boards (protocol number: WCHSIRB-D-2024-183). Informed consent was waived because this retrospective study was based on existing data. All longitudinal clinical records were anonymized. The marking and the measurement of key points on two types of images were completed by two senior dentists after unified training.

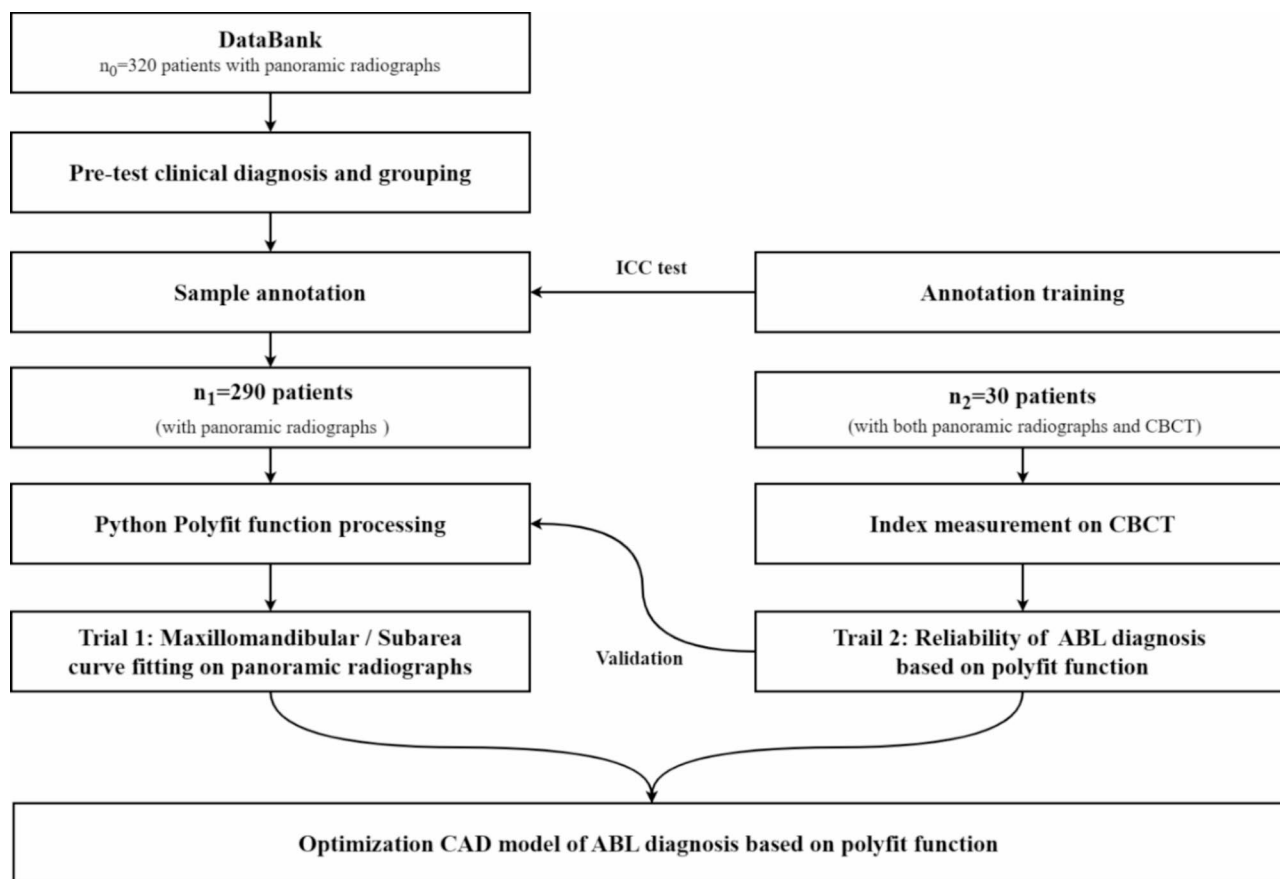


Fig. 1 Study flowchart: the detection to utilize a mathematical approach in alveolar bone resorption evaluation for periodontitis diagnosis on panoramic radiographs via a new annotation rules, fitting function processing and computer aided imaging. 320 patients are included, with 290 panoramic radiographs manually annotated and processed using the polyfit function for statistical analysis, while 30 CBCT scans were measured and compared with the polyfit outputs of the corresponding panoramic radiographs to test reliability. Analysis of the 290 PAN-POL output is conducted and strengthen the reliability and validation of the approach

Clinical diagnosis and eligibility criteria

This retrospective study utilized clinical investigation records of clinical attachment loss that were assessed via direct intraoral examination. The radiograph either PAN or CBCT captured with Guidelines on radiology standards for primary dental care.

The inclusion criteria for this retrospective study required that patients have permanent teeth, as per the FDI tooth numbering system, with at least seven teeth in each quadrant and clear presentation of both teeth and periodontal tissues.

The exclusion criteria targeted severe maxillofacial anomalies such as facial or jaw deformities, and patients with a history of dental implants or removable dentures. Also excluded were cases involving significant gingival swelling, unclear tooth or tissue boundaries, abnormal tooth morphology, severe dentin defects. Additional exclusions covered ABL unrelated to periodontal diseases, traumatic gingival recession, cervical caries, distal clinical attachment level retraction, pus discharge from dental pulp, and vertical root fractures.

According to hospital records, the Departments of Orthodontics and Periodontology had a greater than 10% likelihood of simultaneously obtaining panoramic radiographs and CBCT scans, mostly for periodontal surgery evaluation and orthodontic treatment planning. Hence, for reliability testing specifically, 30 patients who had undergone CBCT containing full dentition within a

three-month window surrounding their Panoramic radiography acquisition were selected.

Panoramic radiography measurements and annotation

The Labelme (version 5.2.1, Anaconda, Austin, TX) annotation tool was used to label panoramic images of 8120 teeth from 290 patients, as displayed in Table 1. It was imperative to assign a distinct identifier to each physiological and anatomical feature of the teeth in question. In accordance with the guidelines set forth by the Fédération Dentaire Internationale (FDI), orientation descriptors included the distal (D), mesial (M), occlusal (O), and gingival (G) surfaces of the teeth on 2D diagnostic images. To streamline the calculation of anatomical teeth height and their attachments, we used specific landmarks depicted in Fig. 2, abbreviated according to these teeth surfaces and regions. The labeling process was halted at the distal aspect of the second molars. In situations where the third molar was missing and recent extraction records confirmed its removal, the analysis exclude the distal aspect of the second molar. This exclusion was taken because of the common presence of localized ABL following an extraction. This exclusion was particularly relevant when annotating the alveolar crest (C) and cemento-enamel junction (J) groups. Conversely, the pulp (T) and root (R) groups extended up to the second molar in each quadrant.

CBCT measurements and annotation

The CBCT data, with a slice thickness of 0.25 mm, were browsed and processed using Mimics (version 21.0, Materialise, Belgium) set to a slice interval of 0.25 mm. Under the guidance of a professional periodontologist, two experienced dentists conducted annotations and measurements following a clockwise chart divided into six sextants according to the distal surface of the canine: starting with the upper right posterior (a, right maxilla) and anterior (b, anterior maxilla), moving to the upper left (c, left maxilla), then to the lower right (f, right mandible), the lower anterior (e, anterior mandible), and finally the lower left (d, left mandible).

Each tooth was initially examined by establishing a cross-sectional view, which was then adjusted to the optimal plane for analysis. The selected image aligned the Y-axis with the long axis of the tooth under examination in both the sagittal and coronal planes, while ensuring that the X-axis of the sagittal plane ran parallel to the tooth's mesiodistal orientation. By utilizing the coronal plane for observation, the ABL for each tooth was meticulously quantified. This process revealed discernible differences between the anterior and posterior teeth, as depicted in Fig. 3(A-C) and Fig. 3(F-H). The ABL for each tooth was measured, also revealing notable differences between anterior in Fig. 3(D-E) and posterior teeth

Table 1 Artificial discrimination of the ABL and detailed grouping according to the segmented regions^a

Characteristic		n(%)
Age, median		13~77, 39
Sex, n (%)		
Male		116, 40%
Female		174, 60%
Alveolar bone non-resorption group (non-ABL)		131, 45%
Alveolar bone resorption group ^a (ABL)		159, 55%
Maxillary regions diagnosed ABL ^b		137, 47%
Maxillary anterior region	b-F ^c	121, 42%
Maxillary posterior region		129, 44%
	a-R	123, 42%
	c-L	112, 39%
Maxillary regions diagnosed ABL		139, 48%
Mandibular anterior region	e-F	137, 47%
Mandibular posterior region		120, 41%
	d-L	116, 40%
	f-R	101, 35%

a. Clinical examination for periodontitis / alveolar bone resorption and overall diagnosis: Based on the retained medical records, confirm whether the patient is diagnosed as periodontitis (or periodontitis has been cured)

b. Examination for segmented diagnosis: Record the site of alveolar bone resorption of the smallest unit in the research (the six sextants, for example, maxillary anterior area, mandibular posterior area) in accordance with the medical records as well as the radiographic secondary diagnosis

c. F: front, R: right, L-left

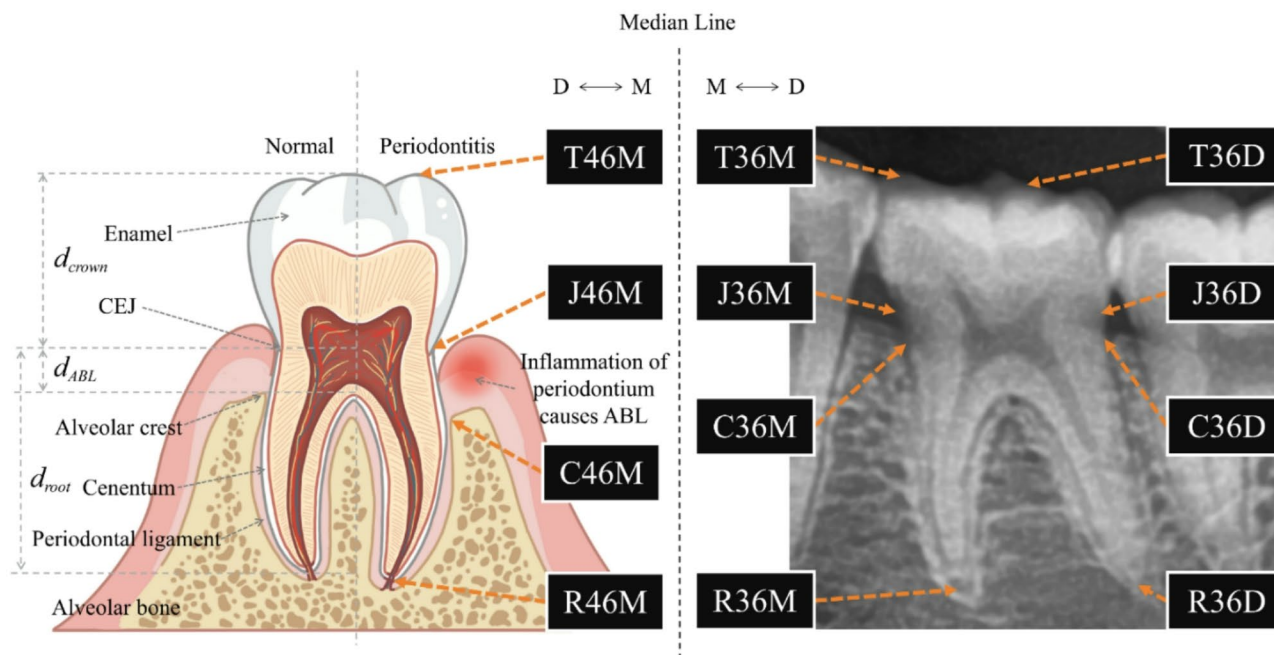


Fig. 2 In the OPG labeling protocol, anatomical landmarks on teeth are marked as follows: 'T' labels (e.g., T36M, T36D) indicate the highest points near the occlusal surfaces; 'R' labels (e.g., R36M, R36D) denote root apices; 'J' labels (e.g., J36M, J36D) highlight the CEJ; and 'C' labels (e.g., C36M, C36D) represent the alveolar crest where the periodontal ligament meets the alveolar bone. These annotations are essential for detailed assessments of dental anatomy and alveolar bone loss, facilitating measurements of tooth and root dimensions and the analysis of alveolar ridge morphology ABL: Alveolar bone loss; CEJ: Cemento-enamel junction; D: Distal surface of tooth; M: Mesial surface of tooth

in Fig. 3(I-J). To ensure consistent measurements across the same patient, each measurement plane was refined to create three distinct sections, with the central section selected to mitigate edge blurring that could compromise accuracy. Besides, all observations were achieved in each CBCT seen directly on a computer screen under appropriate light intensity. All measurements were completed on coronal or sagittal planes that full the screen, using the 400% magnification tool and adjusting image contrast for the clearest structure.

The root length, ABL, and crown length (in millimeters) were each measured four times on both the mesio-distal and buccal-lingual aspects of the teeth. Across all four planes, the root and crown lengths were determined by averaging the two higher values (for posterior teeth, the index derived from the buccal-lingual side was considered acceptable), while ABL was quantified using two absolute linear measurements: the mesiodistal average and the buccal-lingual average, as illustrated in Fig. 3.

It is crucial to note that, when the molars with multiple long roots, C-shaped roots, and curved roots, each root was considered as a singular unit. The length of each root was individually measured, and corresponding absolute and proportionate ABL values were determined. Ultimately, for the target molar, the average ABL of two or three roots was recorded to provide a comprehensive assessment of the tooth's periodontal condition.

Model architecture and computer processing

ABL analysis in PAN-POL model

The PAN-POL model was processed by PyCharm 2023.3.3 (JetBrains, Prague, Czech Republic). For all PANs that completed the labeling of landmarks, the polynomial function "polyfit" was employed to draw a curve that accurately fit all labeled landmarks within each segment.

Considering the morphological characteristics in clinical diagnosis, the curve fitting method is categorized into full-mouth and sectional curve fitting, with the fitting function model limited to a quadratic function as shown in Fig. 4. Each curve was meticulously drawn through points labeled in various groups as described above: the cuspid tip (group T), the cemento-enamel junction (group J), the alveolar crest (group C), and the root apex (group R). According to the best radiological measurements confirmed in reference [37, 38], the key points for binomial fitting are treated as the midpoint of each group of anatomical markers for a single tooth (e.g. the midpoint of the line of J36M and J36D in Fig. 2 is the theoretical neck fitting point). By considering the connecting line on PAN, we aimed to minimize the overall fitting error of the coefficient of determination (R^2). This approach enhances the accuracy and intuitive visualization of both full-mouth and localized ABL. For full-mouth fitting, tooth point coordinate data and a polynomial order are inputted into the polyfit function. The least squares method is then used to calculate the

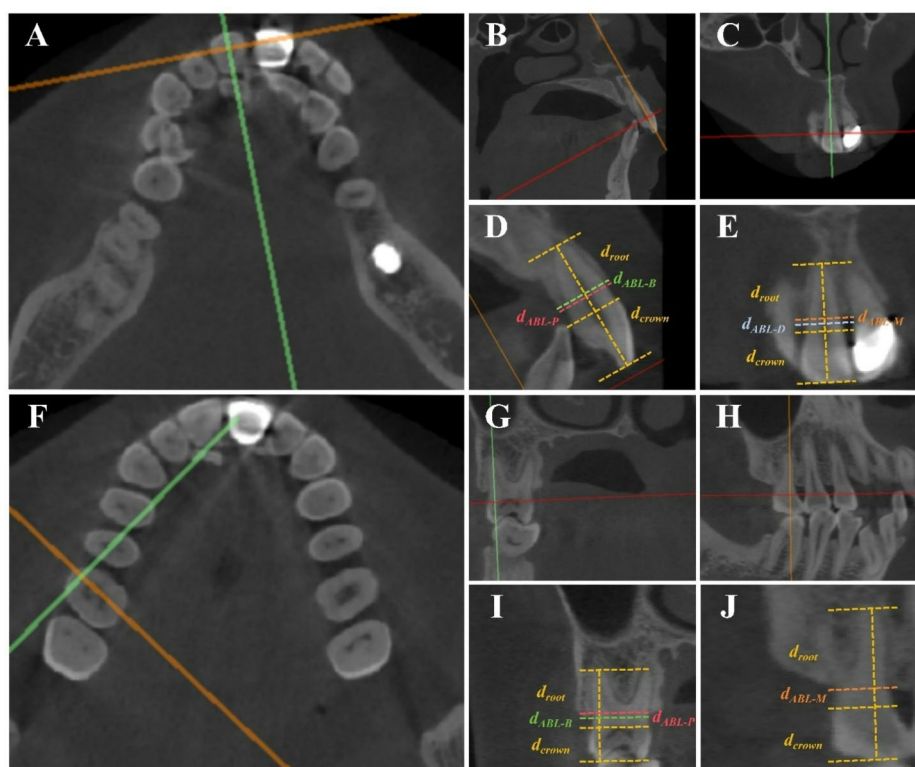


Fig. 3 The measurement of CBCT-related anatomical metrics is indicated. **(A)** Location of the anterior teeth (the central incisor for example) and the premolars. **(B)** The sagittal position was used as the main reference to obtain the ideal image of the tooth root and crown. **(C)** Adjust to obtain the ideal coronal observation plane. **(D)** Corresponding indexes were measured in the plane of Fig. 3B. **(E)** Corresponding indexes were measured respectively in the plane of Fig. 3E. **(F)** Location of the posterior teeth (the first molar for example) with multi-roots. **(G)** Taking mesial root as an example, the optimal coronal observation plane was obtained by adjusting the mesial buccal cusp tip and mesial root, and the longest and clearest images were obtained. **(H)** Obtain the best sagittal position and, if possible, the longest and clearest image of the mesial root. **(I)** Corresponding indexes were measured in the plane of Fig. 3G, especially the length of anatomical tooth roots and the distance from CEJ to alveolar crest (marked as ABL). **(J)** Corresponding indexes were respectively measured in the plane of Fig. 3H. In particular, the length of each root is obtained uniformly in the best measurement plane based on the labial/buccal and lingual/palatal direction of the tooth long axis, whether the mesiodistal measurement can be carried out or not. ABL: Alveolar bone loss; CEJ: Cemento-enamel junction; B: Buccal surface of tooth; P: Palatal surface of tooth; D: Distal surface of tooth; M: Mesial surface of tooth

polynomial coefficients, ensuring the polynomial curve aligns as closely as possible with these data points. From this, we derive three inherent curves for the cuspid tip (r_T), cemento-enamel junction (r_C), and root (r_R), along with one ABL description curve (r_{ABL}) relatively. The final index of these curves is displayed in Table 2. The abscissa at the end of each curve does not necessarily align; in such cases, the remaining three curves are truncated by the abscissa closest to the median line on each curve. Subsequently, the average distance between the curves is calculated.

According to the arrangement of the six quadrants, which include the maxillary and mandibular posterior and anterior regions, segmented curve fitting was employed to accurately reflect local alveolar bone resorption. The vertical distances instead of the Euclidean distances were utilized, since vertical reliability was substantial while horizontal dimensions might have limitations due to magnification errors [39, 40]. Additionally, the values of Crown-root ratio (n_0) and Relative alveolar

bone loss (n_1) were calculated through computer processing of the marked labels, as illustrated in Table 2.

CBCT ABL analysis

Each absolute linear distance index on CBCT ($h_{1\sim3}$) covered multiple tooth positions within each dental sextant, with each tooth position encompassing several measurement sites—each site corresponding to one tooth root in both the buccal-lingual and mesiodistal directions. Patient data collection was directly managed using Microsoft Excel. For each dental region, the mean value of the relative linear distance indices, derived from various measurement planes as noted earlier at each tooth position, was calculated. These mean values were then aggregated with corresponding indices from other teeth within the same region. The software then calculated the average of these combined values, providing a comprehensive assessment for that particular dental region.

Similarly, the full-mouth assessment, which can be viewed as an expanded local region, employed the same

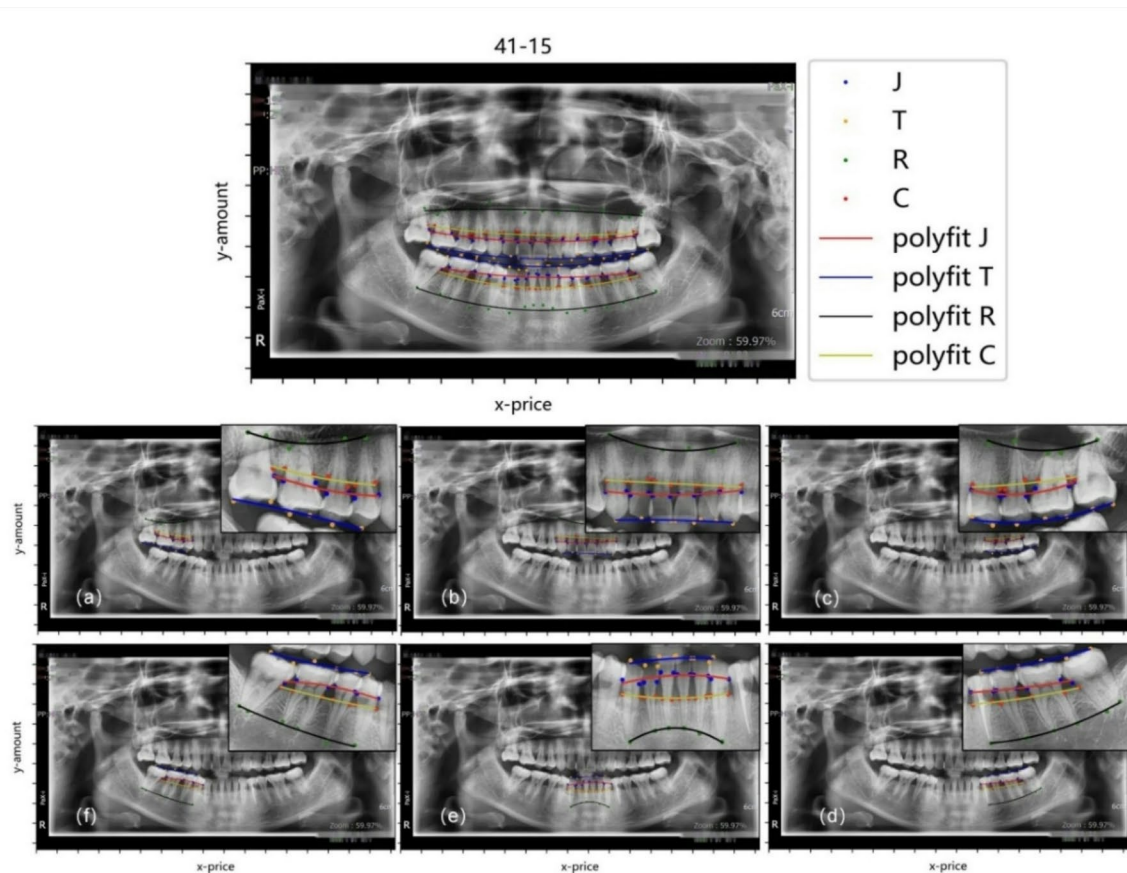


Fig. 4 Computer fitting curves of full-mouth dentition and sectional quadrants

Table 2 The output metrics and respective marker processing on PAN

Metrics	Paraphrase	Measurement reference
Absolute linear distance		
h_1	The vertical distance between the r_T and the r_J in the corresponding section (Crown length).	Fitting curve r_J
h_2	The vertical distance between the r_R and the r_J in the corresponding section (Root length).	
h_3	The vertical distance between the r_C and the r_J in the corresponding section (ABL).	
Relative linear distance		
n_0	Crown-Root Ratio, $n_0 = h_1/h_2$	The vertical distance between the r_R and the r_J in the corresponding section (Root length).
n_1	Relative ABL, $n_1 = h_3/h_2$	
Goodness of fit R^2	In the case of the same abscissa, the consistency between the data points and the approximate fitting curves	

methodology. This systematic and standardized approach to processing the required indicators ensures a thorough analysis of ABL across the entire dentition. It allows for a detailed evaluation of periodontal health and the effectiveness of various treatment modalities on bone structure.

Reliability and validity test

To assess the consistency between different observers, the annotation work was carried out by two senior

dentists. Initially, CBCT data from four patients not involved in the study—two with alveolar bone resorption and two without—were measured. Additionally, the PAN data from 39 not involved randomly selected patients were marked and the data output according to established rules to ensure the randomness of image selection and uniformity of annotation standards. The measurements of ABL (n_1) and the crown-root ratio (n_0) by the two dentists were then compared.

Intraclass correlation efficient test

We employed the Intraclass Correlation Coefficient (ICC) to analyze the two sets of measured or calculated values, using the absolute agreement definition in the two-way random model. The results indicated an ICC range of 0.69 to 1.00 ($P < 0.05$, **Supplement 1** Table 3). Except for the ICC of 0.69 for the ABL in c-segment on the PAN, all other ICC values were greater than 0.75, demonstrating a high level of agreement between the observers for most measurements.

Statistical analysis

Statistical analysis was conducted using data from each patient record, which were simultaneously entered into Microsoft® Excel® 2019 MSO (Microsoft Corp., Redmond, WA) and analyzed using IBM SPSS Statistics 24.0 (IBM, Chicago, IL, USA).

To demonstrate the distinction between the two sets of model-processed data, we compared the full-mouth and segmented ABL(n_1) between the ABL and non-ABL groups, which had been initially classified based on the presence of any ABL as well as ABL in specific segmented regions. And then the proportion indices from 290 manually labeled panoramic images were collected and subjected to a normality test. For groups that demonstrated normal distribution and homogeneity of variance, a single-sample t-test was conducted. Conversely, nonparametric tests were applied to groups exhibiting either non-normal distributions or non-homogeneous variances. After performing a Test of Normality, we used a single-sample t-test or a Mann-Whitney Test to analyze differences in ABL between the non-ABL and ABL groups, as well as the difference between two groups.

Following this, we compared ABL and crown-root ratio (n_0) values from CBCT measurements with those from

the curve-fitted PAN-POL model in 30 patients, aiming to verify the model's reliability. The medians and interquartile ranges were reported, facilitating the evaluation of reference ranges for normal and abnormal values. Subsequent to a further Test of Normality, both the Intraclass Correlation Coefficient (ICC) and the Wilcoxon Signed Ranks Test were employed to assess the reliability of the PAN-POL model in comparison to CBCT measurements.

In addition, R^2 was utilized to gauge how well the curves fit the points and to determine if the curves effectively represented the key point positions for each group. Validation of the model relied on clinical records and was rigorously tested using the R method to ensure its accuracy.

Results

Trial 1: Application of PAN-POL in ABL

Initially, patients were classified into non-ABL and ABL groups based on the presence of any ABL as well as ABL in specific segmented regions, and comparisons were made using the PAN-POL model. This overall diagnosis helped divide analysis involved 159 cases in the ABL group and 131 in the non-ABL group, with findings shown in **Supplement Table 1**. For a more detailed assessment, patients were also categorized within specific quadrants based on the presence of resorption and non-resorption of segmented regions as above.

The initial normality assessment was executed using the Kolmogorov-Smirnov test with Lilliefors correction. Patients with alveolar bone loss, irrespective of the cause of absorption, displayed a significant skew in their ABL data. The majority of the periodontally healthy cohort exhibited a normal distribution, with the exception of group b's non-ABL group ($P = 0.039$). All abnormal ABL

Table 3 Discriminant results of PAN-POL dealing with patients with ABL and non-ABL in each segmented region

Segmented Sectors	M(QR)		Mann-Whitney Test ^a	
	ABL	Non-ABL	Z	P
Full-mouth analysis				
Maxillary region	0.21(0.18, 0.28)	0.14(0.12, 0.16)	-13.19	0.000*
Mandibular region	0.22(0.18, 0.28)	0.14(0.11, 0.16)	-13.29	0.000*
Segmented region analysis				
Maxillary anterior region				
b	0.24(0.20, 0.30)	0.15(0.13, 0.17)	-13.24	0.000*
Maxillary posterior region	0.24(0.20, 0.30)	0.13(0.10, 0.16)	-17.56	0.000*
a	0.24 (0.20, 0.30)	0.14 (0.14, 0.17)	-12.84	0.000*
c	0.22 (0.19, 0.27)	0.12 (0.10, 0.15)	-13.62	0.000*
Mandibular anterior region				
e	0.25 (0.20, 0.33)	0.15 (0.12, 0.17)	-13.45	0.000*
Mandibular posterior region	0.20 (0.17, 0.25)	0.12 (0.10, 0.15)	-17.13	0.000*
d	0.20 (0.18, 0.24)	0.12 (0.10, 0.15)	-13.45	0.000*
f	0.22 (0.19, 0.27)	0.13 (0.11, 0.15)	-13.09	0.000*

M: Median; QR: Interquartile Range a. The intergroup rank-sum test for the data of non-normal distributions *: $P < 0.001$

values ($P=0.000$) were skewed. In the segmented analysis, areas c ($P=0.049$) and b ($P=0.004$) deviated from normality. The integration of frequency histograms with normality curves revealed a pronounced skew in all ABL groups ($P=0.000$ or $P=0.001$), aligning with the epidemiology of periodontitis.

Further segment-specific calculations and comparisons are reported in Table 3 and Supplement Table 2. The Mann-Whitney Test identified a significant statistical disparity in ABL values between pathologically positive and negative groups ($P<0.001$), indicating the model's efficacy in differentiating pathological ABL from physiological resorption.

Comparison of positive patients diagnosed by two different ABL diagnostic methods is shown in Table 4. As expected, ABL values for all segmented diagnoses were higher than those for overall diagnoses, indicating that segmented methods are more accurate and suitable as a sensitive radiographic ABL indicator; except for the overall assessment of lower posterior teeth using both methods ($P=0.077$), statistically significant differences were found in all other groups ($P<0.05$), which may be due to extensive alveolar bone resorption, especially in the mandibular and mandibular central incisor areas. The greatest ABL absorption was observed in the mandibular incisor area, 0.25 (0.20, 0.33) for the segmented method and 0.24 (0.19, 0.33) for the overall evaluation; followed by the segmented diagnosis of the maxillary incisor area, 0.24 (0.20, 0.30), and 0.21 (0.16, 0.27) for the overall assessed ABL group. This comprehensive approach facilitated a precise identification and evaluation of periodontal conditions across various dental regions.

Additionally, the quality of the curve fitting achieved by the PAN-POL model was evaluated, particularly

highlighting the degree of approximation to which the fitting curves conformed. The goodness of fit, indicated by the R^2 value, did not adhere to a normal distribution ($P<0.001$) and was typically close to 1, suggesting a high degree of accuracy in the model's fit for both full-mouth and regional analyses, as shown in Supplement Table 2.

Trial 2: Reliability and accuracy of PAN-POL

The analysis of integrated data from 30 randomly selected PANs and their corresponding CBCT measurements showed no significant differences between the two radiographs for both full-mouth and quadrantal analyses ($P>0.05$), especially in the mandible region, including the left posterior regions of the mandibular (d-segment) and the anterior maxillary region (e-segment) ($P>0.10$), as detailed in Table 5. The results indicated that the ABL measurements obtained by both radiographs were not significantly different, with ICC values ranging from 0.90 to 1.00, suggesting that our model provides reliable and consistent results for assessing ABL.

Additionally, we examined the differences in the measurement of the crown-root ratio. The results showed statistical differences ($P<0.05$, as detailed in Supplement Table 4) between two imaging modalities in all areas except for the mandible region and f-segment, with the ICC values showing no obvious consistency. These findings suggest that there is potential for further refinement and discussion regarding the optimization of crown-root ratio assessments in the current model.

Discussion

In this study, the binary classification of ABL positivity reflects the initial clinical assessment from panoramic radiographs. The application of segmented analysis with

Table 4 Discriminant results of ABL groups of different classification method

Segmented Sectors	M(QR)		Mann-Whitney Test ^a	
	Overall ABL group	Segmented ABL group	Z	P
Full-mouth analysis				
Maxillary region	0.21 (0.17, 0.26)	0.21(0.18, 0.28)	-2.00	0.045*
Mandibular region	0.21 (0.17, 0.28)	0.22(0.18, 0.28)	-1.77	0.077
Segmented region analysis				
Maxillary anterior region				
b	0.21(0.16, 0.27)	0.24(0.20, 0.30)	-3.42	0.001*
Maxillary posterior region	0.20 (0.16, 0.26)	0.23(0.19, 0.30)	-4.55	0.000*
a	0.21 (0.18, 0.28)	0.24(0.20, 0.30)	-3.01	0.003*
c	0.19 (0.15, 0.25)	0.22(0.19, 0.27)	-4.10	0.000*
Mandibular anterior region				
e	0.24 (0.19, 0.33)	0.25(0.20, 0.33)	-2.03	0.042*
Mandibular posterior region	0.18 (0.15, 0.23)	0.20(0.17, 0.25)	-4.15	0.000*
d	0.18 (0.15, 0.22)	0.20(0.18, 0.24)	-3.69	0.000*
f	0.18 (0.15, 0.23)	0.22 (0.19, 0.27)	-4.76	0.000*

M: Median; QR: Interquartile Range a. The intergroup rank-sum test for the data of non-normal distributions *: $P<0.05$

Table 5 Comparison of ABL (n1) in CBCT and PAN

Segmented Sectors	Intraclass Correlation Coefficient, ICC ^a		Wilcoxon Signed Ranks Test ^b	
	Intraclass Correlation ^d	Sig	Z	P
Maxillary region	0.98 ^c	0.000*	-1.51 ^e	0.131
Mandibular region	0.98 ^c	0.000*	-1.68 ^e	0.094
a	0.90 ^c	0.000*	-1.47 ^e	0.141
b	1.00 ^c	0.000*	-1.31 ^e	0.192
c	0.94 ^c	0.000*	-1.47 ^f	0.141
d	0.98 ^c	0.000*	-1.74 ^f	0.082
e	0.96 ^c	0.000*	-1.84 ^f	0.066
f	0.94 ^c	0.000*	-0.90 ^e	0.371

(a) Intraclass Correlation Coefficient Test for the data of abnormal-normal distributions (b) The intergroup rank-sum test for the data of non-normal distributions (c) The estimator is the same, whether the interaction effect is present or not. (d) Type A intraclass correction coefficients using an absolute agreement definition (e) Based on negative ranks (f) Based on positive ranks *: $P<0.05$

computational modeling diminishes the impact of global absorption patterns on local diagnostic judgments, thus refining diagnostic accuracy.

Our data reveal that despite the classification of segmented group patients' ABL as periodontitis in both global and segmented analyses, the widespread mandibular absorption, notably in the anterior region, does not confer statistical significance to the differences between mandibular division methods. Yet, the comparison of specific dental segments shows distinct absorption degree values between the two diagnostic approaches and grouping, highlighting the value of precise segmented assessment in rapid clinical periodontitis diagnosis with panoramic radiographs. For example, in terms of matching the stages of periodontitis and tooth loss, the crown-root ratio is considered a valuable indicator for assessing ABL, yet its role as a standard diagnostic tool remains unclear [41–43]. Consequently, this study aims to develop a mathematical model to calculate this ratio, enabling a reliable and straightforward diagnosis via PAN.

Although image distortion can limit the accuracy of rapid ABL assessment with PAN, this model enhances the utility of PAN in precisely and quantitatively detailing ABL across multiple teeth. This advancement aids in the diagnosis and quantification of periodontal bone loss, offering a more effective tool for clinical evaluations.

In terms of clinical imaging assessment methods, Stramotas et al. [44] suggested that using the crown-root ratio in PANs could circumvent the issues of direct length measurement errors and magnification effects. In addition, research by Hölttä [25], Al-Jamal [45], Yun [46], Wang [47] and Sindi [48] further explored ethnic variations in root-crown ratios for clinical application. Although these studies introduced mathematical ratios to handle absolute linear measurements, they focused on root-crown ratio reference values characterized by geographical and ethnic distribution, overlooking the limitations of single-tooth assessments and individual variability. Previous researchers have also introduced

radiographic indices similar to the crown-root ratio or root-crown ratio for evaluating radiographic alveolar bone levels. Notable examples include the Gingival-Bone Count Index by Dunning and Leach (1960) [49], Periodontitis Severity Index by Adams and Nystrom (1986) [50] and the Bone Loss Index by Teeuw (2009) [51], all of which were developed to quantify and document changes in alveolar bone levels among patients with periodontitis [26]. Building on the use of mathematical methods, studies have improved the precision of measurements like the root-crown ratio across populations. However, they often overlook individual variations and the complexities of single-tooth assessments, limiting their broader applicability [26, 52].

The approach used in this study alternatively enhanced the precision with which the true outline of the bone is approximated, improving the accuracy of diagnostic assessments. The challenge of ABL assessment on PAN due to segment stretch ratios is addressed by using detailed curves and turning points rather than a continuous line. By dividing the dentition into six segments based on imaging and arch turning points, we improved the reliability and of curve fitting relatively. This approach provides a more accurate representation of root lengths and ABL, particularly for teeth prone to bone loss, thus offering a practical method for clinical evaluation.

Our study utilized the PAN-POL model to assess ABL, contrasting it with three-dimensional imaging techniques like those used by Nagao et al. [53]. While our approach incorporates both manual labeling and mathematical modeling, it emphasizes the use of clinical and anatomical knowledge to enhance diagnostic accuracy and relevance. Unlike methods that solely focus on automated image transformation, our model analyzes the entire dental arch, providing a comprehensive overview which is crucial for effective periodontal disease assessment [54, 55].

Our results contrast with prior studies [14–16, 28, 56] that have pointed out the limitations of using PAN for assessing ABL either manually or AI assisted, mainly

attributing our divergence to the application of a mathematical model advanced by the polyfit function and a ratio-based approach. The concept of ratios, specifically by comparing the height of ABL to the length of the tooth root, provides a relative linear parameter indices and takes the place. This method aligns with the techniques used in other studies [26, 57] and allows for a more refined evaluation. Additionally, we utilized the PAN-POL model, a Python-Polyfit-based tool that facilitates polynomial fitting by generating polynomial equations from specified data points. In our study, this function played a crucial role in detecting and quantifying significant changes in alveolar bone across each quadrant and the entire dental arch, and significantly enhancing the accuracy of our assessments.

Despite encouraging results, we recognize the limitations of our study. The sample size might restrict the generalizability of this findings, and the manual annotation process could introduce subjectivity. Future research could explore automated annotation methods with larger samples collection to further and validate the reliability and efficacy of the model, especially when considering the potential confounding factors related to dental treatment. While the model shows promising reliability and consistency comparable to CBCT in its regional evaluations, it does face a challenge from the lack of the analysis module of specific single tooth position and its primary validation on PAN images only. The inclusion of more intuitive clinical investigation results alongside conventional periapical radiographs should be further considered. Since the superimposition of the buccal and lingual cortical plates on 2D imaging was hard to avoid, the comparison between the ABL of single plane on CBCTs and that on PANs could be ulteriorly made. Finally, due to the angular distortions affecting length measurements and vertical bone resorption being overlooked by binomial fitting, some radiological angular and liner markers of localized periodontitis should be further explored to capture such outliers.

In summary, the development of the PAN-POL model provides a more accurate method for assessing ABL on PAN images. By integrating clinical and anatomical knowledge with the application of mathematical approach, our model not only enhances the accuracy of ABL assessment with PAN but also produces results comparable to those from CBCT. Thus, the PAN-POL model has the potential to become a valuable screening tool for periodontal disease diagnosis and oral health monitoring.

Conclusion

In our retrospective study, we created the PAN-POL model to measure ABL on PAN, using clinical and anatomical knowledge. This model accurately differentiates

between resorbed and non-resorbed bone. When compared to CBCT, a widely used diagnostic tool, our model, combined with manual markings on PAN, provided equivalent assessment of ABL with a clearer visual presentation.

We also estimated ABL values and ranges for both periodontitis patients and healthy individuals. The PAN-POL model offers a more accurate and visual way to gauge alveolar bone resorption and could become a valuable screening method, promoting individualized periodontal diagnosis and overall population health assessments.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12903-025-05714-y>.

Supplementary Material 1

Supplementary Material 2

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Author contributions

TE, HJ made the design of the work, managed the project, drafted the work and substantively revised it; TZ, RR and LS collected data. TE, HJ annotated, processed and analyzed the data. TE, HJ and TZ prepared the article; A.A. and LM reviewed and edited the article. XC corrected the manuscript. LJ and HX gave critical guidance and supervision of project as well as financial support. All authors discussed the results and commented on the manuscript. All authors read, discussed and approved the final manuscript.

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Data availability

The pycharm 2023.3.3 software package is available at links: <https://www.jetbrains.com/pycharm/whatsnew/2023-3-3/>. The Anaconda-Labelme 5.2.1 software package is available at links: <https://anaconda.org/conda-forge/labelme>. The markup examples and polyfit codes can be access at github public links: <https://github.com/Claire-JWH/Polyfit-for-ABL>. The image and statistical datasets used and analysed during the current study contained a large amount of patient information, which may not be convenient to disclose. However, they are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The research was in conformity with the guidance of the World Medical Association Helsinki Declaration for biomedical research involving human subjects and the imaging data involved was approved by the institutional review boards (protocol number: WCHSIRB-D-2024-183). Since this retrospective study was based on existing data, informed consent was unnecessary. All longitudinal clinical records were anonymized by the request of the Ethics Committee of Sichuan University.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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